

Cascade direction from a single-cell snapshot: the method, the coupling fix, and where direction reverses

cytokine-MIL project — consolidated progress report
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Abstract

How it works: from one snapshot of perturbed immune cells we give each condition a *discovered* gene signature, then read how much each condition’s cells express *another* condition’s signature — the symmetric part of that cross-engagement tells us *whether* two conditions are coupled, the antisymmetric part (**cross_asym**) tells us *who is upstream*. **What it can do:** recover known cytokine cascades and their direction across five datasets (83–88% directional accuracy) and recover a disease/ vaccination *progression* timeline from a cross-section. **The gate fix:** the coupling-existence test used to over-call ($\sim 77\%$ of pairs “coupled”); a donor-level null plus a hub/degree correction nearly halved the over-call (to 31%) *while raising* known-pair recall ($8/17 \rightarrow 11/17$), and is symmetric so it never touches the direction call. **The reverse finding:** on a T-cell *differentiation* axis the direction call comes out *exactly backwards* — because **cross_asym** assumes the upstream cell *acquires* the downstream program, whereas differentiation is the opposite (the mature cell *retains* the progenitor’s program); a synthetic “tree” control reproduces the reversal and pins it on the acquisition-vs-retention asymmetry, not on topology.

1 The method, exactly (and short)

We never label cells by hand. For every condition X (a cytokine, a stimulus, a state, a time-point) we train a small attention-MIL model to tell “ X vs. control” and read out its **discovered signature** S_X = the top-50 genes by Integrated Gradients through the model. Direction and coupling then both come from one quantity — how much condition a ’s cells express condition b ’s signature, PBS/control-normalised, computed *per cell type* and then aggregated (median across cell types):

$$M[a, b] = s(a, S_b) - s(\text{control}, S_b), \quad s(x, S) = \text{mean expression of gene set } S \text{ in population } x. \quad (1)$$

The matrix M splits into a symmetric and an antisymmetric half, and *that split is the whole method*:

$$\underbrace{\text{coupling}(a, b) = M[a, b] + M[b, a]}_{\text{symmetric} \Rightarrow \text{do they engage each other? (existence)}} \quad \underbrace{\text{cross_asym}(a, b) = M[a, b] - M[b, a]}_{\text{antisymmetric} \Rightarrow \text{who is upstream? (direction)}} \quad (2)$$

Spelled out with the control term, the direction statistic is

$$\text{cross_asym}(a, b) = \underbrace{[s(a, S_b) - s(b, S_a)]}_{\text{raw cross-engagement}} - \underbrace{[\text{ctrl}_{S_b} - \text{ctrl}_{S_a}]}_{\text{control term}}. \quad (3)$$

The biology it encodes: an *upstream* stimulus’s cells carry *both* their own and the (autocrine) *downstream* program, so $s(\text{up}, S_{\text{down}}) > s(\text{down}, S_{\text{up}})$ and **cross_asym** is positive in the upstream-first direction. It is antisymmetric by construction ($\text{cross_asym}(b, a) = -\text{cross_asym}(a, b)$), so its *sign* encodes the arrow. Direction is read *only* on pairs the coupling gate already calls coupled — existence and direction are two different questions.

2 The coupling-gate fix

The gate decides “are a, b coupled?”. The old version **over-called** ($\sim 77\%$ of Oesinghaus pairs “coupled”) for two reasons:

- **Over-power.** The significance null was computed at the *cell* level (thousands of cells), so every mean was razor-sharp and almost everything beat the null. The true unit of independence is the *donor* (~ 10), not the cell.
- **Hub/degree bias.** A broadly-engaged signature (e.g. IL-15, induced by many cytokines) looks coupled to *everything*; this lives in the coupling value itself and survives even a perfect null.

The fix. Replace the cell null with a **donor-level sign-flip null** (unit = donor), and remove hubs by **degree double-centering** of the coupling matrix, $R[i, j] = C[i, j] - d_i - d_j + g$ (d_i = node strength, g = grand mean). The degree correction is symmetric, so it *only* sharpens existence and **never alters direction**. Degree correction was the decisive piece; specificity-only fixes (panel-residualised signatures) did *not* reduce over-call.

Table 1. Coupling gate on Oesinghaus, fix applied step by step.

Gate variant	null unit	over-call	known-pair recall
old (cell-level null)	cells ($\sim 10^3$)	$\sim 77\%$	8/17
+ donor-level null	donors (~ 10)	53%	8/17
+ degree correction	donors	31%	11/17 (2.1$\times$)

Does the trained encoder earn its place? Yes. Holding everything else fixed and swapping *only* how S_X is defined (Integrated Gradients *through the model* vs. plain differential expression, $\times$ two contrasts), DE collapses direction to chance and selects near-disjoint genes (Jaccard= 0.11 with IG). The model’s attributions carry signal a fold-change list does not.

Table 2. Signature ablation on Oesinghaus — the encoder is not removable.

signature	direction acc.	coupled frac.	top hub
IG, vs-PBS (current)	14/17 (82%)	76.5%	IL-15
IG, vs-Panel	14/17 (82%)	77.9%	IL-15
DE, vs-PBS	10/17 (59%)	58.7%	IFN- β
DE, vs-Panel	9/17 (53%)	46.7%	IFN- β

3 What it can and can’t do — across five datasets

All numbers use the **updated method** (signature coupling + donor/degree fix + **cross_asym** direction). Three dataset levers — number of perturbation labels, number of donors, and whether the labels stand in a directional relationship — determine where the method works.

Table 3. Dataset metadata (the levers).

dataset	organism / tissue	cells	donors	cell types	labels	panel
Oesinghaus	human PBMC, 24h	$\sim 9.7\text{M}$	12	~ 18	90 cyt	whole-tx
Sheu	mouse BMDM, 3h	$\sim 295\text{K}$	4*	~ 3	7 stim	500-gene
Imm. Dict. (Cui)	mouse LN, <i>in vivo</i> 4h	$\sim 387\text{K}$	3/cyt	26	86 cyt	whole-tx
Cano-Gamez	human CD4 T	43K	4	2	4 Th	whole-tx
Oelen	human PBMC	$\sim 911\text{K}$	128	9	6 path $\times$ t	whole-tx

*pseudo-donors. Whole-tx datasets reduced to 4000 HVG; coupling used a binary-IG label subset where training cost required it (Oes 24, Cui 12).

Table 4. Results with the updated method.

dataset	coupling over-call	coupling recall	direction (cross_asym)	gate path
Oesinghaus	77 → 31%	11/17 (2.1×)	88% (15/17)	donor+degree
Sheu	→43%	4/5 MUST [†]	86% (6/7)	cell+degree
Imm. Dict.	88 → 44%	6/7	83% (5/6)	cell+degree
Cano-Gamez	100 → 33%	Th17↔iTreg #1	n/a (parallel fates)	cell+degree
Oelen	33 → 47% [‡]	1/3	progression 2/3	donor @128

[†]latent-geometry coupling *failed* 0/2 on Sheu’s targeted panel; signature-space coupling *rescued* 2/2 IFN cascades. [‡]too few labels (6) for degree correction to help.

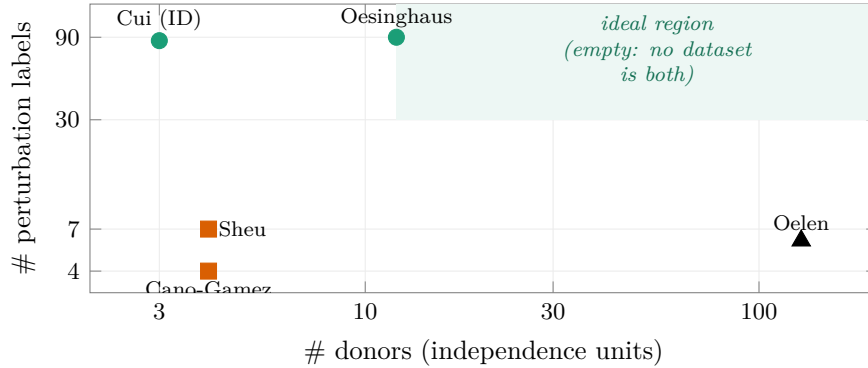


Figure 1. The method needs *both* many donors (powered null) *and* many labels (stable degree correction). **Green:** coupling+direction work; **orange:** partial; **blue:** donor-scale milestone but too few labels. The joint “ideal” corner is empty — the standing gap.

Four metadata-driven rules. (1) *# labels governs degree correction* — works at $\gtrsim 15$ –20 conditions (Oes, Cui), fails at 4–6 (Cano, Oelen). (2) *# donors governs the gate path* — < 8 donors $\Rightarrow$ cell-level fallback; $\geq 8 \Rightarrow$ powered donor-level null (Oelen’s 128 is the scale proof). (3) *Panel breadth governs the shared-activation confound* — a targeted 500-gene panel (Sheu) collapses latent geometry, but signature space (specific genes) is panel-robust. (4) *Label semantics govern whether direction exists* — cytokine perturbations with known cascades give direction (83–88%); parallel fates (Cano) or pathogens (Oelen) give coupling/progression only. *Net:* Oesinghaus uniquely has all three (many labels + ≥ 8 donors + directional perturbations); every other dataset is missing one, exactly where it underperforms.

4 The reverse finding: a recovered timeline but an inverted state order

We pushed the direction statistic onto a **T-cell maturation** axis (naive→effector→memory) on a public SARS-CoV-2 vaccination whole-PBMC CITE-seq atlas (*Nat. Immunol.* 2023; days 0/2/11/28, 6 donors, 61,911 T cells), reusing the disease-progression pipeline verbatim under two framings. **STATE:** condition = {Naive, Effector, Memory} read from CITE *surface protein* (CD45RO/CD27 — RNA-independent, so state labels cannot leak into the RNA signatures), control = day-0 “Resting”. **TIMEPOINT:** condition = {D2, D11, D28}, control = D0. An independent synthetic *apparatus* supplies a distinct-program go/no-go gate.

Table 5. Direction recovery: the apparatus passes, the timeline is recovered, the state order is exactly inverted.

framing	cross_asym acc.	symm. control	Kendall τ	verdict
Apparatus (distinct-program gate)	100%	50%	+1.0	PASS
Timepoint D2<D11<D28	100%	0%	+1.0	AMBER [†]
State Naive<Eff<Mem	0%	100%	-1.0	RED (reversed)

[†]Timepoint is AMBER *only* because the 6-donor bootstrap CI is wide ([0.33, 1.0]); the point estimate is perfect. Recovered orders: timepoint = D2→D11→D28 (truth); state = Memory→Effector→Naive (the exact reverse of truth, hence $\tau = -1.0$).

The tell-tale is the relationship between the direction statistic and its *symmetric* control (**directional_score**). On the apparatus and timepoint axes **cross_asym** $\gg$ control (the healthy cytokine/COVID pattern); on the state axis the two *swap* (Fig. ??) — the synthetic “no-seed” regime, reproduced in real data (Fig. 2).

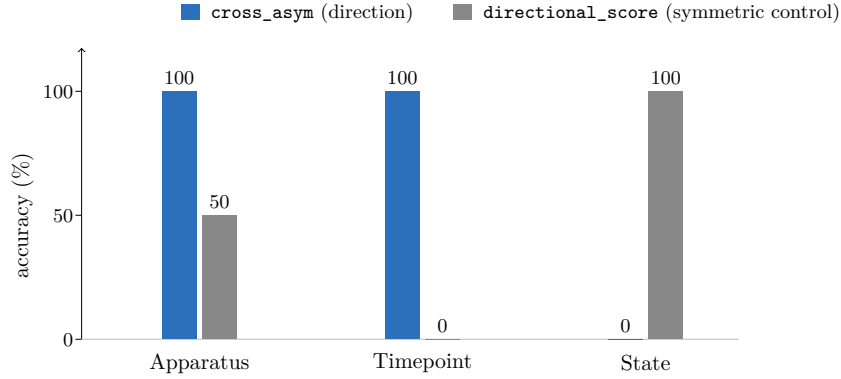


Figure 2. The “inverse fingerprint.” On the apparatus and timepoint axes the antisymmetric direction statistic beats its symmetric control; on the state axis the two swap places — direction 0%, control 100% — the hallmark of an axis where **cross_asym**’s core assumption runs backwards.

We falsified our own first explanation. The obvious guess was “the day-0 control $\approx$ Naive, so the upstream state coincided with the baseline.” A *control-swap decomposition* (holding the signatures fixed, varying only the control term in Eq. (3), including a **ZERO** control = pure cross-engagement) refutes it: the *raw* engagement is the *most* inverted (+0.588 vs +0.140 under Resting), the control *reduces* the inversion, and *no* control choice un-inverts the Naive pairs ($\tau = -1.0$ throughout). The inversion is intrinsic to the axis.

Table 6. Control-swap decomposition — the reversal is control-independent. Correct sign for an X–Naive pair is *negative* (Naive upstream); every control gives *positive*.

control	cross_asym(Eff,Nai)	cross_asym(Mem,Nai)	τ
Resting (original)	+0.107	+0.140	-1.0
ZERO = raw engagement	+0.253	+0.588	-1.0
Balanced (removes naive/mem imbalance)	+0.121	+0.146	-1.0
Naive as control	+0.138	+0.147	-1.0
Memory as control	+0.159	+0.161	-1.0

The mechanism. **cross_asym** assumes program *acquisition*: the upstream cell *gains* the downstream

program (cytokine-autocrine logic). Differentiation is the opposite — a program-*retention* asymmetry: the *mature* cell *retains* the *progenitor’s* program (memory T cells re-express the naive program *IL7R*, *TCF7*, *SELL*, *CCR7*). Hence $s(\text{Memory}, S_{\text{Naive}}) \gg s(\text{Naive}, S_{\text{Memory}})$ and the statistic faithfully points mature→naive. This is not a bug (the apparatus distinct-gate passes) and not fixable by control choice; it is the method meeting an axis whose biology is sign-flipped relative to its assumption.

4.1 A synthetic “tree” confirms: it is acquisition-vs-retention, not topology

To rule out “maybe the chain just has to be linear,” we built a 4th synthetic scenario (*new this session*): a partial-order **tree** $p_0 \rightarrow p_1$, $p_1 \rightarrow p_2$, $p_0 \rightarrow p_3$ where each node carries its own program block *plus half its parent’s full program* (a pure *retention* structure — the descendant carries the ancestor), with *identical* hyperparameters to the three ladder scenarios so the four are directly comparable. The tree reproduces the no-seed/retention regime exactly: **cross_asym** mis-signs *all four* ancestor→descendant edges (each names the descendant as upstream), while the two incomparable sibling pairs stay correctly $\approx$ ambiguous; the reversal magnitude even tracks the carried fraction (direct edges $|\text{cross_asym}| \approx 0.61$; the transitive grandparent edge ≈ 0.26).

Table 7. Four comparable synthetic scenarios (same hyperparameters). A retention structure inverts **cross_asym** whether the topology is a chain or a tree; a distinct-program (acquisition-style) structure is recovered perfectly.

scenario	cross_asym acc.	symm. control	Kendall τ
distinct (acquisition / seeded)	100%	50%	+1.00
monotone_noseed (retention)	0%	50%	−1.00
monotone_seeded (mixed)	67%	50%	+0.33
tree (retention, non-linear)	0%	50%	−1.00

5 Synthesis: an identifiability problem, and how to break it

The reversal is not a defect of the statistic — it is a genuine **identifiability problem**. A single observation $\text{cross_asym}(X, Y) > 0$ (“*X*’s cells carry more of *Y*’s program than vice versa”) is consistent with *two opposite truths*: **(i)** *X* is upstream and is *inducing* *Y*’s program (acquisition — cytokine/timeline case, sign correct), or **(ii)** *Y* is upstream and *X* is *derived from Y* and retains its residual program (retention — differentiation case, sign reversed). The snapshot alone cannot tell these apart. This separates cleanly into how the method should be *used*:

- $|\text{cross_asym}|$ (and coupling = $M + M^\top$) = **coupling strength**: robust and regime-independent. The tree confirms it — it fired confidently on every real edge and correctly abstained (≈ 0) on unrelated siblings.
- $\text{sign}(\text{cross_asym})$ = **direction**: interpretable only once the *regime* (acquisition vs. retention) is fixed. Confidently right under acquisition, confidently *wrong* under retention.

Fixing the regime needs an **arrow from outside the snapshot** — exactly the classic trajectory-inference identifiability issue (why RNA velocity needs a spliced/unspliced clock). The practical disambiguators: **(a) time or a perturbation** sets the arrow directly — which is *why the timepoint framing worked and the state framing did not* (the vaccine imposes t_0 ; a pure cross-section of one lineage has no arrow); **(b) a known root / applied stimulus** (the cytokine case — the perturbation *is* the upstream); **(c) lineage constraints** (two different cell types cannot be related by differentiation, so a shared signature there is more likely acquisition); **(d) the gene-class of the carried program** — induction tends to leave the downstream’s *inducible/response* genes in the upstream, whereas retention leaves the upstream’s *identity/TF* genes in the downstream, a potential *internal* discriminator we have not yet tested.

Bottom line for the project. The coupling gate is fixed and validated; the direction statistic is validated on acquisition-type axes (cytokine cascades 83–88%; vaccination/ disease progression timelines). We have now *mapped its boundary*: on cell-state differentiation axes it is a faithful *coupling* detector but a sign-flipped *direction* detector, and we understand precisely why. The next milestones follow directly: an acquisition-vs-retention discriminator (the gene-class probe, or a forward-seed/backward-carry synthetic pair), a donor-level *direction* null for novel-pair claims, and a dataset that is simultaneously high-donor *and* many-label *and* directional — the empty “ideal” corner of Figure 1.